UNIVERSITY OF CALGARY

Endorsed, or Just Enforced? Personality and Preferences for Online Learning During COVID-19

by

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A THESIS

SUBMITTED TO THE FACULTY OF GRADUATE STUDIES

IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE

DEGREE OF MASTER OF SCIENCE

GRADUATE PROGRAM IN PSYCHOLOGY

CALGARY, ALBERTA

SEPTEMBER, 2022

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Abstract

Research on online learning and remote work has historically been conducted using individuals who were already attracted to those modalities. However, COVID-19 forced many organizations operating in-person to suddenly switch to online formats, creating a novel population of individuals who were not necessarily attracted to or experienced in online learning or remote work. This also created the opportunity to learn more about how individual differences influence perceptions of online modalities. This cross-sectional study investigated the relationships between university students' (N = 465) personalities, preferences for online learning, and preferences for remote work. We found a modest overall effect of HEXACO personality on preferences for online learning and remote work. Individuals high in Honesty-Humility or Extraversion are less likely to prefer online modalities. More students preferred in-person learning than online learning, and those who preferred in-person learning felt more strongly about their decision on average than those who preferred online learning. The number of students who preferred remote work was approximately equal to those who preferred in-person work, but hybrid work was most popular. Additionally, an EFA of factors of the learning experience revealed six most important mechanisms of online learning preference, which will inform further scale development. The findings of this study suggest that personality traits affect how individuals experience online learning differently, and predict preferences for virtual modalities, which may have implications for course drop-out rates and work recruitment. Understanding this relationship can guide the design of online learning and remote work programs to attract and accommodate a wider range of individuals, or individuals with desired traits, based on experiences and attitudes that shape their preferences.

Keywords: personality, online learning, remote work, preferences, HEXACO, COVID-19

Preface

This thesis is original, unpublished, independent work by the author, Kaylee Ann MacLean. The study reported in this thesis was covered by Ethics Certificate number REB21-1968, issued by the University of Calgary Conjoint Health Ethics Board for the project "Personality and Perceptions of Online Learning" on March 2, 2022.

Acknowledgements

To my supervisor Dr. Kibeom Lee for being the reason I get to be in graduate school at the University of Calgary, and whose guidance, patience, and kindness made the completion of this thesis project possible;

To my parents Susan and Bradley MacLean for instilling in me the value of education and my own potential, and for helping make my goals possible;

To Erika Lieu and Jonathan Henke for warmly welcoming me into our cohort and being great people to call my colleagues and friends;

To Chelsie Hart and Alexandra Debusschere for being supportive friends before I even moved;

To Matthew Dawson for encouraging and supporting me academically and mentally;

To Ann and Charlie Rogers for ensuring I had family and a piece of home nearby;

To my baby niece Emerson for brightening my day when I need it;

To my friends Leanne Funk, Kimberly Crann, and Loshini Vickneshwaran for being one in a million and making Calgary home;

To my first year TAs Brittany Lindsay and Emiko Muraki who had their work cut out for them but got me through statistics with dedication and grace;

To my professors Drs. Kibeom Lee, Tom O'Neill, Joshua Bourdage, and Derek Chapman who make me excited to learn in this program and field;

To Jia Bin for generously helping with qualitative data analysis;

To my committee members Drs. Joshua Bourdage, Laura Hambley, and Piers Steel for their help in shaping and improving my project;

To Drs. Stephen Colbert and Ben Zalkind for helping me be the best version of myself;

Thank you for being the shoulders of giants that I stand on.

Dedication

For Sammie, who kept me company through late nights of studying and was always by my side.

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Introduction

Research on online learning and remote work has historically been conducted using populations who were already willingly signed up for those modalities. Whether that is remote work (Campbell, Eley, & McAllister, 2013; O'Neill, Hambley, & Chatellier, 2014; O'Neill, Hambley, Greidanus, MacDonnell, & Kline, 2009) or online learning (Cohen & Baruth, 2017; Dikaya, Avanesian, Dikiy, Kirik, & Egorova, 2021; Nemanich, Banks, & Vera, 2009; Paechter & Maier, 2010; Phelan, 2012; Song, Singleton, Hill, & Koh, 2004), the populations available to be studied were predominantly those who were already drawn to those positions for one reason or another. Without a population with both the equal opportunity for and the experience with online modalities, we were only ever able to rigorously answer the question, "who *likes* learning and working online?" However, in the years since March 2020, many educational institutions and workplaces around the world were forced online in response to the risk of the COVID-19 pandemic. People who were used to going to work in person or learning in a classroom found themselves adapting, with varying satisfaction and success, to a complete shift in their daily routine. The sudden switch to virtual modalities created a population that did not exist on a large scale before. There are now millions of people with experience learning and working online who were not necessarily attracted to those modalities.

While past research has been conducted on those who voluntarily opted for online learning and remote work, the aforementioned population of those who involuntarily switched online presented a different opportunity. As such, the present research project aimed to understand not only who likes learning and working online, but who *dislikes* learning and working online?

Part 1 of this project determines what patterns exist between the personalities and preferences of students who have done online learning, as well as their future intentions to choose to work remotely. This is done through measurement of students' (1) personalities, (2) attitudes toward online learning, and (3) preferences for future remote work. The existence of any patterns among the three variables provides useful insight into what types of students are more likely to favor or disfavor online learning and remote work.

To further progress our understanding of student attitudes towards online learning, Part 2 of this study elucidates the mechanisms of the relationship between students' experiences with online learning and their overall preference for online learning. Through an exploratory factor analysis (EFA) of factors of the online learning experience, the higher-order dimensions of the online learning experience are identified for the purpose of developing a future scale that will more directly investigate *why* students favor or disfavor online learning. Such a scale will provide an even more detailed framework for educational institutions to understand how students' overall attitudes toward online learning are formed, and identify opportunities for improvement in the online learning experience.

Part 1

Personality and Online Learning

Online learning enrollment rates have been steadily increasing over the past decade (Seaman, 2017) and skyrocketed following the onset of the COVID-19 pandemic. In the fall of 2020, 75% (11.8 million) of undergraduate students in the United States participated in at least one online course and 44% (7 million) undergraduate students participated in exclusively online courses ((*The NCES Fast Facts Tool provides quick answers to many education questions (National Center for Education Statistics)*, n.d.). These rates were up from 6 million students

enrolled in at least one online course and 2.4 million enrolled exclusively in online courses just one year prior in 2019, representing a 97% and 186% increase, respectively (National Center for Education Statistics, 2022).

However, dropout rates for online courses have historically been higher than traditional in-person courses (Atchley, Wingenbach, & Akers, 2013). This suggests that there is a disconnect between the degree to which online learning is becoming more common within higher educational institutions and the degree to which online learning is being received by students.

To address retention issues in online learning, it is essential to understand who is attracted or averse to online learning in the first place. Historically, most studies on online learning were conducted on students who had already willingly signed up for online courses, and may have been biased in favor of online learning, at least initially. This has stunted our ability to measure attitudes toward online learning among students who may not necessarily be attracted to it in the first place, and the characteristics of those students. The mass exodus to online learning and remote work during COVID-19 created a greater opportunity and need for research on online learning. We can now sample learners who had experience with online learning and who dislike it, not only those who were biased toward it by virtue of self-selecting into that mode. This study took advantage of the timely opportunity to understand not only the individual differences of people who are attracted to and prefer online learning, but also the traits of students who are averse to this method.

Personality is one measure of individual differences that has received much attention in the literature to predict a variety of academic outcomes in online learning, such as final course grade and dropout rate (Meredith, 2011), satisfaction in online and blended learning (Bolliger &

Erichsen, 2013), participation in online courses (Furnham, Trickey, & Hyde, 2012), Massive Open Online Course ("MOOC"; e.g. Coursera/Udema/edX) completion intention (Gupta & Misra, 2021), trust in online learning environments (Bhagat, Wu, & Chang, 2019), willingness and anxiety toward distance learning (Randler, Horzum, & Vollmer, 2014), perceived value for money of online courses (Watjatrakul, 2020), and online engagement during COVID-19 (Quigley, Bradley, Playfoot, & Harrad, 2022). The number and breadth of studies on personality and academic outcomes in online learning make it clear that personality has demonstrated relevance and utility in understanding students' online learning experiences.

However, previous studies on personality and online learning have yielded a variety of results, even on similar criterions, and have demonstrated a lack of consensus. For instance, although (Topi, Valacich, & Rao, 2002) concluded that introverts were less satisfied with online communications than extraverts, in a study by Pavalache-Ille and Cocorada (Pavalache-Ilie & Cocorada, 2014), high introversion was associated with participation in and preference for online learning. In contrast, in Cohen and Baruth's (2017) study, Extraversion was not a significant predictor of satisfaction with online courses, and (Audet, Levine, Metin, Koestner, & Barcan, 2021) found none of the FFM traits predicted satisfaction with online courses. Most recently, (Quigley et al., 2022) found that Extraversion predicted both participation and performance in online courses.

Quigley et al. (2022) also found that Agreeableness was unrelated to engagement in online courses, and Openness to Experience was unrelated to all forms of engagement except for emotional engagement. This result differed from previous findings where Openness to Experience was strongly related to engagement in online courses (Audet et al., 2021; Mammadov, 2022).

Neuroticism has also been examined in studies such as Bhagat et al. (2019), where Neuroticism had a significant negative effect on perceptions toward online learning. However, Gupta (2021) did not replicate that result, and found no significant effect of Neuroticism on intentions to complete Massive Open Online Courses (MOOCs), yet a significant effect of the other four out of five FFM traits on intention to complete MOOCs.

Conscientiousness is a highly consistent predictor of outcomes in the academic setting such as academic performance, and Cohen and Baruth (2017) found that it also significantly predicted students' satisfaction with online courses. However, Smidt (Smidt, 2015) found that conscientious college students are also more satisfied with in-person courses.

Therefore, on its own, the finding of Cohen and Baruth (2017) is not particularly useful in understanding students' perceptions and informing online learning program design. Their results describe a link between personality and course satisfaction, but it is not necessarily specific to online learning. That is, it could well be that highly conscientious students are more satisfied with their courses than students lower in conscientiousness, regardless of mode of delivery. This highlights the importance of comparing attitudes toward online learning to in-person learning, rather than investigating online modes in isolation, before drawing conclusions about how to shape online learning programs.

In fact, none of the aforementioned studies measured both online and in-person modes when evaluating the role of personality in online learning. Audet et al. (2021) investigated students' personalities, motivation, and engagement in online courses during COVID-19, but did not measure how this differed from students' in-person experiences. Therefore, these findings are not necessarily specific to online learning. Similarly, Bhagat et al. (2019) found correlations between personality traits and perceptions of online learning that may not be limited to online

learning. Highly agreeable students were more likely to agree with statements describing the instructional design of their courses like, “ I think sharing knowledge through online discussions is a good idea,” but agreeable students are already more likely to give positive course evaluations anyway, even in in-person courses (Patrick, 2011), so we cannot conclude that Agreeableness predicts attitudes toward online learning that are different from attitudes toward in-person learning. Without the ability to contextualize how online learning may differ from in-person learning, we are not able to fully understand the role of personality in predicting attitudes toward online learning. This study built upon the contributions of past studies by having participants compare their experiences in both online and in-person learning when reporting their perceptions, which is crucial to developing evidence-driven recommendations for designing online learning programs.

Another feature of the present research is the adoption of the six-dimensional structure of personality, known as the HEXACO model of personality structure (Ashton & Lee, 2007). While past studies primarily relied on the Five-Factor Model (“FFM”) of personality, the HEXACO has been shown to not only have high validity but also provide a richer profile of personality than five-factor models through its inclusion of a sixth factor, Honesty-Humility (Ashton & Lee, 2007). Honesty-Humility is particularly relevant in the context of online learning and will provide a deeper understanding of online learning preferences by yielding information not otherwise captured by five-factor models. Specifically, the H factor of HEXACO will allow for the investigation of how personality relates to the desire to cheat, which is especially a concern with online learning. Studies that used the Five-Factor Model of personality without Honesty-Humility have so far not been able to capture this important aspect of the online learning experience.

The HEXACO differs from the FFM in a few other ways, such as by not including the FFM trait of Neuroticism. Instead, some aspects of Neuroticism are covered by HEXACO's (low) Agreeableness, such as whether someone has a quick temper, and others by Emotionality, such as how easily someone becomes stressed. These differences do not exist in name alone and are demonstrated by statistical differences. A review by (Anglim, Horwood, Smillie, Marrero, & Wood, 2020) revealed that, compared to HEXACO, the FFM had greater variance in observed correlations, consistent with the greater variability found in the FFM questionnaires themselves. In fact, despite sharing a name, the traits of Agreeableness in the FFM were often as closely related to HEXACO's Honesty-Humility as they were to HEXACO's Agreeableness. Additionally, the FFM's Neuroticism was also closely related to HEXACO's Extraversion and Emotionality.

Finally, in light of the significant context of the pandemic on this research, there is a need for contemporary research on how the personality trait Emotionality correlates with online learning preferences. Emotionality (comprised of Fearfulness, Anxiety, Dependence, and Sentimentality within the HEXACO) presents a challenge to accurately measure due to the extenuating circumstances of the COVID-19 pandemic. For instance, (Clabaugh, Duque, & Fields, 2021) found that students who had recently been forced into online learning during the COVID-19 pandemic experienced elevated levels of stress, uncertainty, and overall harm to emotional well-being. This comes as no surprise due to the highly disruptive effect COVID-19 had on students' daily lives early in the pandemic. Therefore, more research is necessary to more accurately determine the role between Emotionality and students' perceptions of online learning. Special attention was given in the study to prompt students to do their best to think about their experiences with their most recent two semesters of online learning, rather than their initial

semesters of online learning at the beginning of the COVID-19 pandemic. If focus were given to online learning at the beginning of the pandemic, the initial adjustment period of unexpectedly switching formats may have evoked emotions that are not reflective of more stable attitudes toward online learning. Despite the aforementioned concerns, this study nonetheless advances our understanding of how Emotionality, and all of the other HEXACO personality traits, correlates with online learning preferences following the onset of COVID-19.

Using multiple regression, the relationship between students' HEXACO personality traits and overall preference for online learning was evaluated. The results of this analysis represents the first step in understanding how students' individual differences affect their online learning preference. With further research, this information can be used to guide higher educational institutions in designing online learning programs that do not repel prospective students based on qualities that are unrelated to the primary learning outcomes.

Personality and Remote Work

In addition to studying personality and online learning preference, Part 1 of this study also investigated the relationship between personality and remote work preference. Remote work, also called working from home, telework, and telecommuting, is similar to online learning in that an individual completes their work outside of the traditional office space and does not need to physically commute to start their day (Olson, 1983). Whereas online learning can be asynchronous, synchronous, or involve a hybrid with some in person days, remote work exists on a spectrum from fully remote (never going into an office) to a hybrid[1] arrangement of spending some days remote and some days in an office.

[1] "Hybrid work," also called live mobile work, activity based work, etc., can be considered a distinct mode of a combination of in-person and remote work, but since the conditions of this mode are not well-defined in previous statistics, it was not directly examined in this study.

Before COVID-19, remote work was expanding modestly, with 25% of wage and salary workers at least occasionally working from home ("29 percent of wage and salary workers could work at home in their primary job in 2017–18 : The Economics Daily: U.S. Bureau of Labor Statistics," 2019), up from 10% seven years prior (Mateyka, Rapino, & Landivar, 2012). After the onset of the pandemic in 2020, that proportion grew to make up more than 40% of the U.S. workforce (Bloom, n.d.). It is unclear how "occasionally working from home" is defined, which further punctuates how limited our understanding is of remote work. While remote work pre-COVID was performed by fewer people, often inconsistently, remote work as we understand it today has expanded to become more common and more formally structured.

Despite the proportion of remote workers more than doubling, overall attitudes towards remote work are mixed. For some workers, remote work afforded them the flexibility to control their own schedule and environment, overcome accessibility challenges, and reach job opportunities that were previously far away. For others, the challenges of remote work have overshadowed the benefits.

Before the pandemic forced millions of workers to abruptly change their daily life, remote work on average had a lower job satisfaction compared to traditional in-person work (Smith, Patmos, & Pitts, 2018). However, that effect became less clear over the course of the pandemic. (Kondratowicz & Godlewska-Werner, 2021) found that remote work during the COVID-19 was associated with high job and work satisfaction; however, this finding was not compared to job and life satisfaction with in-person work. Additionally, (Bellmann & Hübler, 2020) found that the introduction of remote work improved job satisfaction only initially, and work-life balance eventually suffered.

Personality did not predict turnover intention among remote workers during the pandemic (Evans, Meyers, Van De Calseyde, & Stavrova, 2022) but it did play a role in certain job outcomes. Extraversion and Conscientiousness positively predicted job outcomes early on but were later associated with decreased satisfaction and engagement, while employees low on Extraversion and Conscientiousness experienced improvement in those outcomes (Evans et al., 2022). Workers high on Emotionality experienced greater burnout at the beginning of the remote work shift but this effect too became more variable over time (Evans et al., 2022).

Overall, the effect of personality on remote work preference is still a blind spot in the literature, even more so than personality and online learning preference. Like with online learning, many past studies tended to investigate the effect of personality on remote work outcomes, such as performance, satisfaction, or turnover intentions, without also measuring perceptions of in-person work (Campbell et al., 2013; Evans et al., 2022; Smith et al., 2018). Without the ability to compare remote and in-person work, it is more difficult to assert that the observed effects of personality on job outcomes are specific to remote work itself.

By comparing the effect of personality on both online learning and remote work, the findings of the present study sheds some light on how similarly individual differences affect preference for both online learning and prospective remote work. If the relationship between personality and preference follows the same pattern for both online learning and remote work, that would suggest that there may be a “type” of individual that is most compatible with not only online learning, but with remote work as well. Understanding that pattern is key to the ability for organizations that employ remote workers to design accommodations and attract candidates with desired traits.

Preferences for Online Learning and Remote Work

A final aim of Part 1 of this study is to investigate the connection between students' overall preferences for online learning and their preferences for future remote work. Many of the common complaints about remote work mimic those about online learning, which may suggest that attitudes toward remote work and online learning are generally similar. Challenges such as lack of face-to-face communication (Smith et al., 2018), weaker interpersonal relationships, and difficulties fostering teamwork (Zhang, Yu, & Marin, 2021) plague remote work as well as online learning. While both promise increased flexibility, that autonomy comes at a high cost of self-discipline and demand for internal motivation.

Considering some of the similar characteristics of both remote work and online learning, the analysis employed by the present study elucidates the extent to which students' preferences for future remote work are similar to their preference for online learning. Since this is a cross-sectional study, no conclusions can be drawn about the effect of online learning experiences with future job type choice; however, some insight can be gained into the degree to which the trend of students' preferences for learning modality correlates with their preferences for prospective remote work. The degree to which the two are related may further punctuate the importance of thoughtful design and implementation of online learning. Students' experiences with such programs may have far-reaching effects beyond their school years as they make their emergence into the workforce. For example, if a student has a negative experience with online learning and expects remote work to be similar to online learning, they may be averse to future job opportunities to work remotely. Therefore, online learning course design has the potential to influence trends of attraction to remote work in the future, and by extension, the supply of satisfied workers to fill openings in that modality. In light of the trend of increasing remote work,

significantly accelerated by COVID-19, it is all the more important to understand the extent to which students' perceptions of this format is being shaped early on.

Part 2

Mechanisms of Online Learning Experiences and Preference

Adopting a thematically exploratory approach in this study served to honor the context of significant transformation that surrounds and necessitates this research. This approach is most salient in Part 2 of the study, which utilized an Exploratory Factor Analysis (EFA) to expand the breadth of knowledge that can be concluded from students' evaluations of online learning. It is appropriate to return to the basics of assessment creation rather than rely on assessments that have already been truncated or have been grandfathered from times when technological advancement and relevance looked quite different than today. Consider again the fact that previous samples of online learners were typically conveniently sampled and were already willing to do online learning. Recycling the assessments developed using those samples may result in missing facets of the online learning experience that may have been nonsignificant to those participants, but may in fact be highly relevant to participants who are more averse to online learning.

An EFA of students' evaluations of experiences of online learning elucidated which dimensions of the learning experience are most relevant to students' preference of learning modality. This step provided the most insight into the mechanisms behind students' preferences for online learning, which will be used in a future study to create a validated, formal assessment of students' attitudes toward online learning. Higher educational institutions will then be able to use such assessment to gauge student attitudes toward existing program designs, understand barriers and perks of the online learning program, and consider changes accordingly.

This study took several steps to contribute to a more rigorous repertoire of knowledge of individual differences in online learning, with the overall aim to guide higher educational institutions in implementing online learning programs more effectively. First, more so serendipitously, this study took advantage of the opportunity to study a less biased sample of students who were forced into online learning without necessarily being attracted to it. Second, this study also attempted to address consistency and replicability issues in the literature by utilizing the HEXACO model of personality assessment rather than the more variable Five Factor Model of personality (or other less popular, and in some cases less valid personality models). Third and finally, this study meets a need to measure perceptions of online learning compared to in-person learning.

Ultimately, this study aimed to enhance our current understanding of students' personality and preference for online learning, analyze the link between students' preferences for online learning and their preference for future remote work, and illustrate the relationship between personality, preference for online learning, and preference for future remote work. Higher educational institutions can anticipate the trend of increasing online course enrollment to continue even beyond the mandates of COVID-19. Some of these institutions have so far turned to popular personality myths and ineffective online readiness surveys to guide their understanding of their prospective students' compatibility with online courses, to no great success (Wladis & Samuels, 2016). With online learning and remote work rapidly becoming more mainstream, it is more important now than ever for the rate of empirical research to catch up to the rate of expansion in these sectors.

Current Research

In this study, we evaluated the relationship between personality traits, overall preference for online learning, and preference for future remote work with an undergraduate student sample. Additionally, we evaluated students' attitudes toward aspects of their learning experience to identify key aspects that contribute positively to the preference of online learning over in person learning (or vice versa). Additionally, the present research provides a conceptual basis for the development of a survey in a follow-up study to determine the factors that drive students' preference for online learning. We examined these issues using a university undergraduate student sample that has experienced both online and in person learning without the undue influence of self-selection one way or the other. Overall, this study answers the following research questions:

Research Question 1 (RQ1): To what extent does overall preference for online learning correspond to preference for future remote work?

Research Question 2 (RQ2): What is the relationship between the HEXACO personality factors, overall preference for online learning, and preference for future remote work?

Research Question 3 (RQ3): Which dimensions of the learning experience are most relevant in determining overall preference for online learning?

Methods

This study adopted a largely exploratory approach to answering the above questions, and did not posit hypotheses. There are several reasons for this decision. First, almost no past studies have directly measured the influence of personality on online learning preference. Given that one of the key arguments for the necessity of this study is that preference cannot be conflated with satisfaction, impression, or other measures of attitudes toward online learning, those measures were deemed insufficient reference points for hypothesizing about preference. Instead, an exploratory approach was deemed appropriate for this study due to the novelty of the study's key contributions. Additionally, the COVID-19 pandemic is a new and significant context under which past studies were not conducted. The resulting shift online caused an exponential expansion of the population of individuals who were experiencing online learning for the first time, and who had never been studied before. I believe it would have been naïve to assume that the conclusions of past studies would not come into question in light of those circumstances. However, the very first research was still being conducted regarding how COVID-19 was impacting learning. Several relevant studies have been published during the execution of this research study that did not exist at the conception of this project, so it was not yet clear how those changes were manifesting. The findings of some of those studies have been incorporated into the introduction to make thesis as informative as possible, but it would have inappropriate to retroactively use those findings to inform hypotheses. In the absence of explicit hypotheses, this study did offer several a priori plans of analysis.

First, to investigate RQ1, the correlation between overall preference for online learning and preference for future remote work was to be examined. Next, to investigate RQ2, two multiple regression analyses were planned - one to determine the relationship of HEXACO

personality traits and overall preference for online learning, and one to determine the relationship of HEXACO personality traits and preference for future remote work. Finally, to investigate RQ3, an EFA was planned to determine which dimensions of the learning experience are most relevant in determining overall preference for online learning. This EFA serves to provide the basis for scale development in the future, so that an even more detailed online learning experience survey can be created to answer questions not yet answered by this study.

Participants

Participants were recruited through the University of Calgary's Research Participation System (RPS), hosted and managed by the cloud-based software, Sona Systems. University of Calgary undergraduates are able to browse this system and sign up for studies they are eligible for. Additionally, students of participating Psychology courses may receive bonus course credit for completing studies, worth up to 2% of their final course grade. This study was listed on RPS for participants to complete the survey in exchange for 1.0 credit. There were no eligibility criteria for this study that excluded any student on any basis; any student who had the ability to access the RPS system and had internet access was eligible to participate. The study remained available for up to 500 sign-ups on a first-come, first-served basis. There were no additional incentives offered for completing the study.

Data Cleaning

In total, 486 students submitted the survey out of the 500 students who signed up. Low quality data was screened based on missing data, response time, and attention check response, in that order (DeSimone & Harms, 2018). Out of the 486 participants, one participant's data was deleted due to a lack of response to all three sections of the survey. All other participants had almost no missing data for any questions in survey Parts 1-3.

Next, response time was analyzed to screen out data that was submitted too quickly and therefore unlikely to have been thoughtful and effortful responses. A logical cutoff point of faster than 4 second per item (less than 8 minutes and 30 seconds for the entire questionnaire) was used to account for reading both the items and the instructions in each section. This cutoff point was based on the response time of a non-participant who was instructed to read and answer the questions randomly and completed the survey in 10 minutes. Thirty-five participants completed the survey in under 10 minutes, but participants were only flagged if they completed the survey 1.5 minutes faster than the non-participant, to account for individual differences in reading speed. Therefore, 21 participants were flagged for unreasonably fast response time.

Finally, the 21 flagged participants were screened for whether they passed the attention check in Part 3 of the survey. While the participant with the longest response time out of the 21 did pass the attention check, very few of the other flagged participants passed the attention check, so the data of 20 participants were deleted. The attention check was not used as a standalone method to screen out any other participants.

Ultimately, the data of 21 participants were deleted from the dataset for low quality data. While their data was not used in subsequent quantitative analyses, their qualitative responses in Section 3b were preserved for analysis since they are reasonably thoughtful and effortful responses.

Sample Demographics

The final sample of 465 participants ranged from 17 to 48 years old, with an average age of 20 and median age of 19. Participants who identify as women made up the majority of the participant pool at 79%, followed by 19% men, 1% gender variant/non-conforming, and 0.4% preferred not to answer. See Table 1.

Table 1

Sample Demographics

Variable		Sample (*N* = 465)	
		n	***% Or M (SD)***
Age		454	20 (3.15)
Gender	Woman	369	79.4%
	Man	89	19.1%
	Gender variant/non-conforming	5	1.1%
	Prefer not to answer	2	0.4%
Sex	Female	372	80.0%
	Male	92	19.8%
	Prefer not to disclose	1	0.2%

Note. 11 participants did not provide their age.

Materials

The survey was administered using the third-party website Qualtrics, which participants were redirected to after signing up for the study on RPS. The survey was anonymized and did not collect any personally identifying information, whether directly (e.g. asking for their name, email, etc.) or indirectly (e.g. recording IP addresses, location, etc.)

The survey consisted of three main sections: a personality inventory, an original scale measuring attitudes toward online learning, and an original scale measuring attitudes toward remote work. A fourth section measuring additional demographic and relevant academic, work, and personal information was also included at the end. (See Appendix A for the full survey).

Section 1: Personality

Ashton and Lee's(Ashton & Lee, 2009) 60-item HEXACO-PI-R was used to measure the personalities of the participants. The use of the 60-item inventory instead of the standard

100-item inventory served to keep the overall survey length as brief as possible, thereby maximizing the number of total responses while preserving the integrity of the test. Participants were prompted with the instruction, “Below you will find a series of statements about you. Please read each statement and decide how much you agree or disagree with that statement. Then select your response using the options to the right of each statement. Please answer every statement, even if you are not completely sure of your response.” They were then presented with 60 statements such as, “I would be quite bored by a visit to an art gallery,” and “I plan ahead and organize things, to avoid scrambling at the last minute,” and asked to rate how much each statement described them on a scale of 1 (strongly disagree) to 5 (strongly agree).

Section 2: Online Learning

The second section of the survey, which measures factors influencing students’ overall preference for online work, was developed based on scales found in existing literature (Cohen & Baruth, 2017; Haddon & Lewis, 1994; Paechter & Maier, 2010; Song et al., 2004) and personal insight of what I considered to be logical inclusions or extensions. In the first section, Section 2a, participants were prompted with the instruction, “The following list includes the pros and cons of online learning. Please indicate how you would evaluate your online learning experience with respect to each of these items compared to in-person learning. That is, for each of the following items, to what extent online learning is worse or better than in-person learning? Indicate your response using the following scale.” They were then presented with a list of 40 aspects of learning, such as “Work/life balance, “ and “Opportunity to make new friends.” The participants rated the items on a scale of -3 (Online learning is much worse) to 3 (Online learning is much better).

This section also includes an item (Section 2b) measuring overall preference for online learning: “Overall, which form of learning do you prefer more: online learning or in-person learning?” on a scale of -3 (I prefer online learning much more) to 3 (I prefer in-person learning much more).

Section 3: Remote Work

The final section was dedicated to gauging students’ interest in future hypothetical remote work employment opportunities. This section was inspired in part by (Keller & Karau, 2013) “Online Course Impressions Scale” and Grippaldi’s (2002) “Attitudes” section of their survey of telecommuting attitudes among government finance professionals. Because there are many terms related to remote work that are used interchangeably, such as telework, telecommuting, and working from home, a brief definition of remote work was provided in the instructions.

Participants were prompted with the instruction, “Now imagine you are considering full-time work after University for a position that could be done either remotely or in-person. The following statements refer to your preferences for remote work and in-person work. ‘Remote work’ refers to the practice of working online, outside of a traditional in-person office space. One might work from their computer at home or from a secondary location like a library, café, or personally rented office space. Note that remote work is on a continuum, and can range from fully remote (spending 100% of your working time away from a traditional office space) to occasionally remote (spending most of your time at a traditional office space). For the purposes of this survey, assume that remote work means 80% or more of your time is spent working remotely. For the following statements, indicate how much you agree or disagree with that statement.” Participants were then presented with 10 statements regarding their attitudes toward

remote work such as “Overall, the positives of remote work outweigh the negatives,” and “Remote work might be good for some people but not for me,” which they rated on a scale of 1 (strongly disagree) to 5 (strongly agree). Question 1 of this section served as an attention check.

This section also included a final open-ended question asking students to disclose factors that are most important to them when deciding whether to take a job in remote work. While this question is not included in the quantitative analysis of this study, it was opportunity to gather data for exploratory factor analysis that might explain some of the factors behind students’ preference to hypothetically pursue remote work, and inform future studies that attempt to more closely focus on that topic.

Section 4: Additional Items

Finally, an additional section gathering demographic and contextual information was included at the end of the survey. Section 4 measured participants’ age, gender identity, and sex, COVID-related anxiety (the extent to which personal fear about COVID plays a role in course selection decisions), significant personal factors (the extent to which factors such as financial strain, extenuating family obligations, parenthood, etc. plays a role in course selection decisions), and self-rated academic performance in most recent online courses compared to most recent in person courses.[2]

Procedure

After signing up for the study on RPS, all participants were provided with a link to the same exact survey to be completed at their convenience. Participants were presented with a consent form upon accessing the link, which described the purpose and procedure of the study, assured anonymity of responses, and informed them of the minor risks of psychological distress

[2] Additional variables related to employment and academic history were gathered for the purposes of contextualizing the sample and can be found in Appendix A.

while participating in the study due to the questions about COVID-19 at the end of the survey. Additionally, resources for the Campus Wellness Centre were provided in the event that any participant would like to seek psychological support. Participants were informed that proceeding with the survey beyond the consent form would be interpreted as consent to participate in the study, but that they could withdraw at any time with no penalty. They then proceeded through the survey in the order presented above, at their convenience, and with no time limit. They could also go back and change their responses as they wished until they submitted their survey. Participants were provided with a final consent statement to agree to have their responses included in the study just before submitting their survey. After submitting, they were automatically redirected to the RPS website and assigned 1.0 research study credit.

Results

To answer the research questions posited by this study, all quantitative data were analyzed using IBM SPSS Statistics 28, excluding Section 3b, which gathered open-ended responses and was analyzed for latent content using the qualitative analysis software Quirkos.

RQ1: To what extent does overall preference for online learning correspond to preference for future remote work?

Descriptive Statistics and Correlations

Table 2

Means, Standard Deviations, and Correlations of the Study Variables

Variable	*M*	*SD*	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1. Honesty-Humility	3.37	0.66	-													
2. Emotionality	3.53	0.66	0.09*	-												
3. Extraversion	3.06	0.68	-0.06	-.24**	-											
4. Agreeableness	3.19	0.62	0.34**	-0.04	0.02	-										
5. Conscientiousness	3.54	0.64	0.18**	.18**	.14**	0.00	-									
6. Openness to Experience	3.49	0.63	0.10*	0.05	0.04	-0.05	0.04	-								
7. Online learning preference	-0.55	1.94	-.15**	-0.07	-.17**	0.00	-.13**	0.03	-							
8. Remote work preference	0.02	0.97	-.18**	-.08*	-.22**	-0.01	-.14**	.08*	.55**	-						
9. Age	19.98	3.16	.13**	-.15**	0.07	0.03	0.08	.15**	.12*	.12*	-					
10. Gender	1.26	0.67	-0.07	-.14**	-0.03	0.01	-.12*	0.07	0.02	0.09	0.01	-				
11. Sex	1.21	0.48	-.10*	-.21**	0.06	0.07	-.13**	-0.02	0.01	0.07	0.01	.72**	-			
12. COVID anxiety	2.06	0.98	0.02	.17**	-0.22**	0.00	-0.02	0.06	.20**	.24**	0.00	0.03	-0.02	-		
13. Personal factors	2.35	1.15	-0.08	0.05	-0.13**	-0.07	0.01	.11*	.17**	.13**	.14**	0.06	0.02	.36**	-	
14. Academic performance	4.36	1.64	-.12*	-0.04	-0.10*	0.06	0.09	-0.01	.47**	.28**	.10*	-0.02	-0.05	0.03	0.02	-

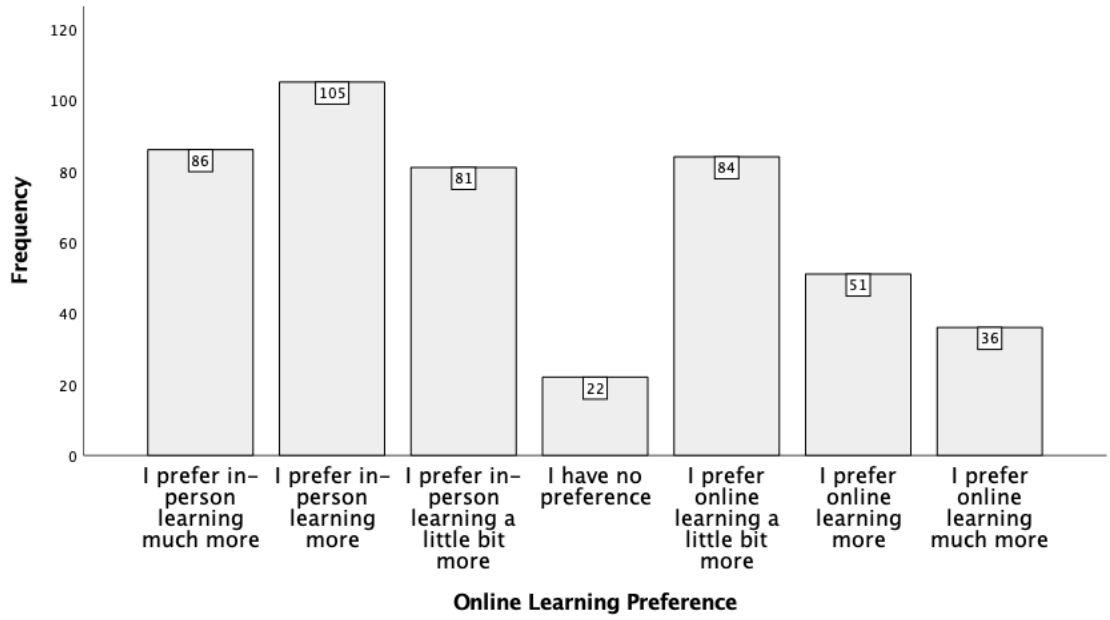
* $p < 0.05$ ** $p < 0.01$

Descriptive statistics and correlation matrices are shown in Table 2. One of the first observations that stands out in the descriptive statistics in Table 2 is the slightly negative mean for online learning preference. Since this variable was measured on a 7-scale from -3 (“I prefer in-person learning much more”) to 3 (“I prefer online learning much more”), a slightly negative

mean indicates that the student participants generally preferred in-person learning more than online learning. This finding is not entirely consistent with Dikaya et al(Dikaya et al., 2021) who found that, among students experiencing online learning during the COVID-19 pandemic, the number of positive and negative overall ratings of online learning were approximately equal. The mean overall preference for online learning among participants in the current study was indeed close to zero, but the median value for students' online learning preference was -1, and the mode was -2. Therefore, the mean alone does not capture the finding that not only were there more participants who preferred in-person learning, the average participant who preferred in-person learning did so more strongly than the average participant who preferred online learning. See Figure 1 for the frequency and spread of participants' responses.

Figure 1

Frequency of Responses to Overall Preference for Online Learning



Participants' preference for remote work, however, was more equally split and centered around a mean of zero, indicating that there is no consensus among participants of remote work being more or less preferred. Among the nine questions measuring attitudes toward future remote work (See Part 3 in Appendix A), four questions measured preference for remote work, two questions measured preference for hybrid work, and three questions measured preference for in-person work. For each student, their mean response value to the questions in each of the three groups was calculated to create variables of average remote work preference, hybrid work preference, and in-person work preference. The averages of participants' remote and in-person work preference were normally distributed, but the average of participants' hybrid preference was negatively skewed (-.48), indicating a stronger preference for a hybrid work model than fully remote or in-person work. See Table 3 for a breakdown of participants' preference for remote, hybrid, and in-person work.

Table 3

Preference for Remote, Hybrid, and In-person Work

	Mean	Median	Mode
Remote Work	.02	.00	.50
Hybrid Work	.53	.50	1.00
In-person Work	-.13	.00	.33

Table 2 reveals several findings about the relationship between personality and preference for online learning. First, the correlations between the HEXACO factors and overall preference for online learning are generally weak. However, four factors do reach statistical

significance. Honesty-Humility (r = -.15, p < .001), Extraversion (r = -.17, p < .001), and Conscientiousness (r = -.13, p = .003) were all negatively associated with online learning preference at the .01 level, and Emotionality (r = -.080, p = .069) was negatively associated with online learning preference at the .05 level. A similar pattern was observed with respect to remote work preference; Honesty-Humility (r = -.18, p < .001), Extraversion (r = -.22, p < .001), Conscientiousness (r = -.146, p = .002) were significant at the .01 level, and Emotionality (r = -.090, p = .036) and Openness to Experience (r = -.08, p = .037) were significant at the .05 level. Openness to Experience was significantly correlated with preference for remote work but not preference for online learning. This could be due to the fact that most participants had never experienced remote work, so their Openness to Experience would have an impact on how comfortable or willing they are to try a new work modality in the future. All participants have already experienced online learning though, so their attitudes toward it are likely more strongly based on other factors besides their comfort or willingness to a new learning modality.

The most significant finding, and the focus of RQ1, was the strong association between preference for online learning and preference for future remote work. Students' preference for online learning is strongly correlated (r = .55, p < .001) with their preference for future remote work. Therefore, students who are generally attracted to online learning are much more likely to be attracted to future opportunities of remote work.

Additional variables describing the sample further contextualized the relationship between personality and the outcome variables. Age, gender, and sex reached significance with a few HEXACO traits and preference for online learning and remote work. In particular, older participants were slightly more likely to prefer online learning and remote work compared to younger participants. This finding highlights the potential benefits of online modes for

individuals who may be balancing childcare or other responsibilities, or who may be working and taking coursework at the same time and value the flexibility of online modes. Overall, however, age, gender, and sex together accounted for relatively little variance in online learning preference when learning preference was regressed on the three variables (R^2_{adj} = .01).

Unsurprisingly, participants who reported higher consideration of COVID-19 or extenuating personal factors in selecting their coursework were more likely to favor online learning and remote work. This illustrates the "double-edged sword" of the timing of this study. On the one hand, COVID-19 made this sample exist, but on the other hand, it underscores the larger context of uncertainty and change during the shift to online learning. Indeed, concern and consideration of COVID-19 itself is a variable that influences preference outside of individual differences such as personality.

Another notable finding is the strong association between participants' subjective evaluation of their academic performance and their preference for online learning. Students who considered their academic performance to have been better during online learning compared to in-person learning were more likely to favor online learning. While academic performance, even subjectively, was not the focus of this study, it is clear that it plays an important role in students' attitudes toward online learning.

Research Question 2 further investigates the contribution of students' personalities in predicting their preference for online learning and remote work.

RQ2: What is the relationship between the HEXACO personality factors, overall preference for online learning, and preference for future remote work?

Multiple Linear Regression: Personality and Online Learning

Two multiple regression analyses were performed to determine how the HEXACO personality variables contribute to the prediction of online learning preference and of future remote work preference.

First, online learning preference was regressed on the six HEXACO personality variables (Table 4). The assumption of normality did not appear to be violated; the residual values were bimodally distributed, however, the mean was centered around 0. The assumptions of linearity and homoscedasticity also did not appear to be violated. Additionally, multicollinearity was not a concern; no tolerance value was less than 0.1 and no VIF exceeded 10 for any of the variables.

A significant regression equation was found ($F(6, 458) = 5.80$, $p < .001$), with an R^2 of .071 and $R^2_{adjusted} = .058$. A student's HEXACO personality profile significantly predicts their overall preference for online learning. Individually, Honesty-Humility ($b = -.48$, $p = .001$) and Extraversion ($b = -.55$, $p < .001$) were significant predictors of overall preference for online learning. Emotionality almost reached significance as a predictor of overall preference for online learning ($b = -.27$, $p = .058$), but the lower and upper bounds of the confidence interval for b include zero. Agreeableness ($b = .18$, $p = .234$), Conscientiousness ($b = -.17$, $p = .240$), and Openness to Experience ($b = 0.18$, $p = .192$) were not significant predictors of overall preference for online learning.

Table 4

Multiple regression of HEXACO personality traits and overall preference for online learning

Unstandardized Coefficients		Standardized Coefficients			95% CI	
b	*SE*	*Beta*	*t*	*p*	Lower Bound	Upper Bound

H	-0.48	0.15	-0.16	-3.30	.001	-0.77	-0.20
E	-0.27	0.14	-0.09	-1.90	.058	-0.54	0.01
X	-0.55	0.14	-0.19	-4.06	<.001	-0.82	-0.29
A	0.18	0.15	0.06	1.19	.234	-0.12	0.48
C	-0.17	0.15	-0.06	-1.18	.24	-0.45	0.11
O	0.18	0.14	0.06	1.31	.192	-0.09	0.46

Multiple Linear Regression: Personality and Remote Work

Next, preference for future remote work was regressed on the same set of predictor variables (see Table 5). A significant regression equation was found ($F(6, 458) = 10.519$, $p < .001$), with an R^2 of .121 and $R^2_{adjusted} = .110$. A student's HEXACO personality profile also significantly predicts their overall preference for remote work. Individually, Honesty-Humility ($b = -.32$, $p < .001$), Emotionality ($b = -.18$, $p = .009$), Extraversion ($b = -.37$, $p < .001$), and Openness to Experience ($b = .19$, $p = .005$), were significant predictors of preference for future remote work. The higher a student scored in Honesty-Humility, Emotionality, or Extraversion, the more likely they were to prefer working in-person over remote work, whereas the higher a student scored in Openness to Experience, the more likely they were to prefer remote work over in-person work. Figure 2 illustrates the standardized Beta coefficients of the HEXACO personality factors that significantly predict online learning and remote work preferences.

Table 5

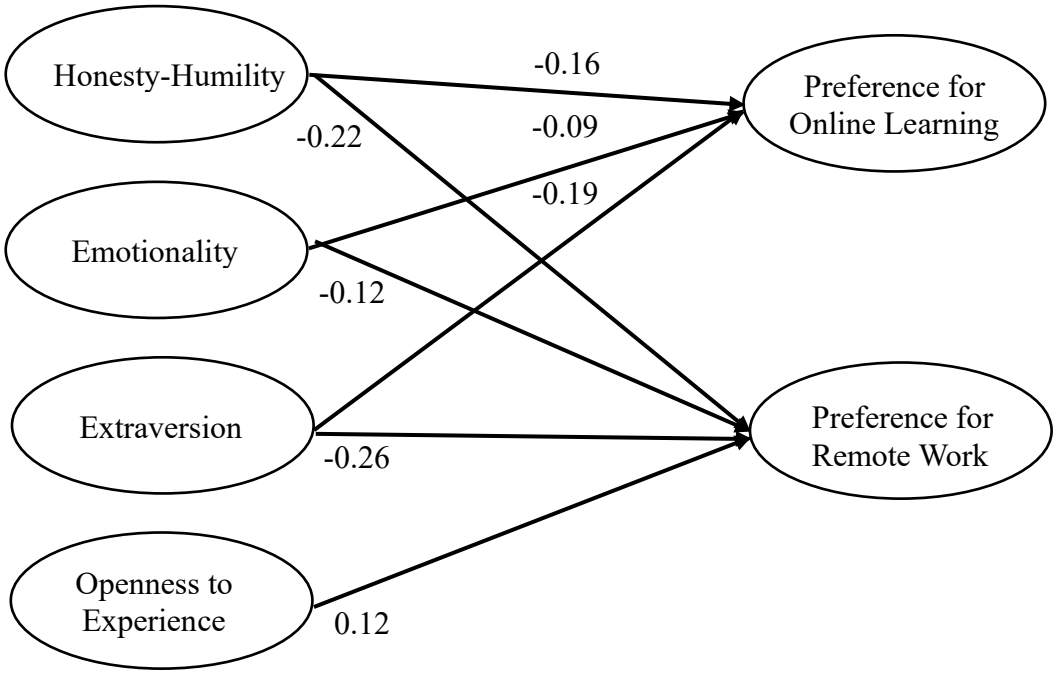
Multiple regression of HEXACO personality traits and preference for future remote work

Unstandardized Coefficients	Standardized Coefficients		95% CI

	b	*SE*	*Beta*	*t*	*p*	Lower Bound	Upper Bound
H	-0.32	0.07	-0.22	-4.5	<.001	-0.46	-0.18
E	-0.18	0.07	-0.12	-2.62	.009	-0.31	-0.04
X	-0.37	0.07	-0.26	-5.58	<.001	-0.5	-0.24
A	0.11	0.07	0.07	1.52	.129	-0.03	0.26
C	-0.07	0.07	-0.04	-0.93	.355	-0.2	0.07
O	0.19	0.07	0.12	2.82	.005	0.06	0.33

Figure 2

Summary of Multiple Regression Analyses



Overall, both multiple regressions were statistically significant and show that personality does play some role in influencing preference toward both online learning and future remote work. Several HEXACO personality traits predicted preference; Honesty-Humility, Emotionality, and Extraversion all significantly predicted both online learning and remote work

preferences. In general, the predictive value of personality was slightly stronger for remote work than for online learning preference. This is corroborated by the fact that Openness to Experience significantly predicted remote work, but not online learning preference. Agreeableness and Conscientiousness did not predict either outcome variable. This is particularly interesting since the correlations shown in Table 2 indicated that Conscientiousness was significantly correlated with both outcome variables, and Openness to Experience was not significantly correlated with either outcome variable.

To further understand the most important mechanisms of preference for remote work, an EFA was performed in the a priori analysis.

RQ3: Which dimensions of the learning experience are most relevant in determining overall preference for online learning?

Exploratory Factor Analysis (EFA)

To investigate RQ3, an exploratory factor analysis was performed to determine which dimensions of the learning experience are most relevant in determining overall preference for online learning. This analysis was conducted using the responses to the 40 items in Section 2a of the survey. These items described specific parts of the learning experience, such as “Work/life balance,” and “Opportunity to make new friends,” which participants rated as being better or worse in online learning compared to in-person learning. The goal of this analysis was to reveal the structure of the relationship between the items in this section of the survey and the outcome variable.

The EFA was conducted via a principal components analysis using a Varimax[3] rotation with Kaiser normalization. A parallel analysis using the macro PROCESS (Hayes, 2013) supported the extraction of 6 variables, and we decided to extract and rotate the first 6 factors.

Analysis of the 6 extracted factors yielded insight into the statistical groupings of the survey items, shown in Table 6. The extracted factors represent dimensions of students' learning experiences that may differ depending on the mode of learning. The 6 extracted factors are summarized in Table 6, and the EFA matrix is presented in Table 7.

Table 6

Extracted Factor Themes and Descriptions

	Theme	Description
1	Social Engagement	Experience of engaging with other students and professors inside and outside of the classroom
2	Industriousness	Personal qualities and behaviors to produce high-quality work
3	Resources	Having the resources and accommodations, both tangible and intangible, to complete coursework
4	Counterproductivity	Ease and temptation to engage in counterproductive behaviors
5	Individual Participation	Ease and satisfaction with class activities completed individually, such as presenting
6	Life Outside of School	Ability to customize the time and resources dedicated to life outside of school

[3] A Promax rotation was also applied to this data and yielded factors that could logically be defined with the same groupings of items as those found with the Varimax rotation.

Table 7

Rotated Factor Matrix

	Factor					
	1	2	3	4	5	6
Social Activities	**0.80**	-0.03	-0.04	0.15	-0.04	0.08
Making Friends	**0.79**	-0.02	-0.06	0.17	-0.02	0.06
Extracurriculars	**0.74**	0.00	0.00	0.16	0.00	0.13
Community	**0.69**	0.04	-0.09	0.07	0.06	0.10
Rapport	**0.66**	0.08	0.06	0.16	0.10	-0.02
Collaboration	**0.65**	0.16	0.11	0.00	0.10	0.06
Communication with Students	**0.57**	0.29	0.14	0.01	0.03	0.09
Engaging Lectures	**0.56**	0.27	0.07	-0.12	0.14	-0.08
Financial Value	**0.53**	0.24	-0.08	0.01	0.16	0.17
Class Discussions	**0.50**	0.27	-0.01	-0.05	0.10	-0.04
Communicating with Instructors	**0.47**	0.29	0.23	-0.20	0.13	-0.11
Engagingness as Audience	**0.44**	0.36	0.19	-0.12	0.19	-0.10
Standards Quality	-0.03	**0.68**	0.10	0.21	0.04	-0.12
Actual Quality	0.06	**0.67**	0.02	0.25	0.07	-0.04
Feeling of Accomplishment in Course	0.15	**0.64**	0.21	0.01	0.10	0.23
Feeling of Accomplishment In Items	0.08	**0.63**	0.15	0.03	0.05	0.23
Self-Discipline	0.34	**0.61**	0.00	0.15	-0.01	0.07
Self-Motivation	0.38	**0.57**	-0.03	0.17	-0.05	0.16
Overall Amount of Learning	0.37	**0.50**	0.02	-0.02	0.00	-0.02
Psychological Health	0.15	**0.42**	0.08	0.01	0.20	0.27
Physical Health	0.20	**0.35**	0.02	0.17	0.21	0.19

Access to Tech Resources	-0.05	-0.01	**0.71**	0.05	-0.10	0.15
Access to Accommodations	-0.07	0.03	**0.56**	0.09	0.11	-0.05
Resolution of Tech Difficulties	0.17	0.08	**0.56**	0.07	0.00	0.11
Getting Questions Answered	0.02	0.28	**0.48**	-0.16	0.10	-0.07
Turnaround Time for Grading	0.00	0.02	**0.41**	0.05	0.12	0.06
Avoiding Temptation to Cheat	0.20	0.22	-0.04	**0.59**	0.18	-0.04
Avoiding Temptation to Cheat in Group	0.08	0.18	0.35	**0.54**	-0.01	-0.09
Ease of Doing Bare Minimum	0.02	0.24	0.03	**0.52**	-0.09	0.03
Ease of Cheating in Group	-0.08	-0.14	0.11	**0.49**	0.23	0.22
Ease of Cheating	0.15	0.10	0.48	**0.48**	0.01	-0.03
Attendance	0.17	0.34	-0.06	**0.37**	0.08	-0.09
Ease of Presenting	0.06	-0.01	0.07	0.01	**0.72**	0.11
Comfort with Presenting	0.09	0.08	0.09	0.11	**0.66**	0.06
Class Participation	0.26	0.12	0.02	-0.03	**0.52**	-0.05
Evaluation Fairness	0.02	0.26	0.13	0.21	**0.35**	-0.06
Amount of Free Time	0.10	0.11	0.05	-0.12	-0.03	**0.59**
Schedule Flexibility	-0.02	0.03	0.08	0.03	-0.07	**0.59**
Work Life Balance	0.17	0.14	0.03	0.07	0.05	**0.53**
Financial Savings	-0.01	-0.01	-0.01	0.01	0.12	**0.49**

Note. Extraction method: Principal Axis Factoring. Rotation Method: Varimax with Kaiser Normalization. Bolded values indicate the factors that the respective items most appropriately load onto.

Additionally, a two-tailed t-test was completed to analyze the correlation between each extracted factor and overall preference for online learning. Factor 1 (Social Engagement), Factor

2 (Industriousness), and Factor 6 (Life Outside of School) significantly correlated with overall preference for online learning, suggesting that the experiences encapsulated by those factors are most influential to students' overall preference for online learning. Factor 3 (Resources), Factor 4 (Counterproductivity), and Factor 5 (Individual Participation) did not significantly correlate with overall preference for online learning and the experiences encapsulated by those factors are not as influential to students' overall preference for online learning (see Table 8).

Table 8

Correlation of Extracted Factors and Overall Preference of Online Learning

	r	*p*
Factor 1	-.53***	<.001
Factor 2	-.19***	<.001
Factor 3	0.04	.439
Factor 4	0.04	.39
Factor 5	-0.05	.239
Factor 6	-.32***	<.001

*** $p < .001$

Of note in the above factor loadings is the fact that the item measuring "financial value," measured by the item "Overall financial value (i.e. getting "the most for your money" from your courses)," loaded onto Factor 1: Social Engagement. This is interesting in that it may suggest that social engagement is key to making students feel like they are getting their money's worth

from their learning experience. In the future scale[4], this item will be used as a dependent variable (like "overall preference for online learning") to further investigate this idea.

Finally, the Pearson correlations between the extracted factors and HEXACO traits were analyzed and are presented in Table 9. Factor 1 (Social Engagement) was significantly negatively correlated with participants' Extraversion, and interestingly, Conscientiousness. This indicates that students higher in Honesty-Humility or Conscientiousness are especially likely to believe that online learning is worse for social engagement. Factor 2 (Industriousness) and Factor 3 (Resources) were not significantly correlated with any personality traits. Factor 4 (Counterproductivity) was positively correlated with Honesty and Conscientiousness, and negatively correlated with Emotionality. Students higher in Honesty-Humility or Conscientiousness are especially likely to believe that online learning is more conducive to their personal counterproductivity, but students higher in Emotionality are slightly likely to believe the opposite. As it is widely believed that cheating is easier and more common in online learning, it makes sense that more conscientious and honest students would be able to realize and admit that fact. Factor 5 (Individual Participation) was weakly correlated with Honesty-Humility, and Factor 6 (Life Outside of School) was not significantly correlated with any personality traits.

Table 9

Correlation of Extracted Factors with HEXACO

	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6
H	-0.02	0.04	0.06	.15**	-.10*	-0.08
E	-0.04	-0.01	-0.03	-.10*	-0.04	0.02
X	-.21**	-0.07	0.00	0.05	-0.02	-0.05
A	0.00	0.09	-0.01	0.05	-0.03	-0.02

[4] See Appendix C for plan for scale development.

C	-.15**	0.02	0.04	.15**	-0.05	-0.01
O	-0.02	0.03	-0.06	0.01	0.02	-0.06

* $p < 0.05$ ** $p < 0.01$

This EFA will be used to inform the future creation of a reliable assessment to measure attitudes toward online learning. The extracted factors shown in Table 6 will guide the writing of 4-6 survey items related to each theme. These items will describe specific experiences much like Part 2 of the survey used in the present study (e.g. “Opportunity for socializing during class” may be used to represent Factor 1). The items will be compiled into a questionnaire that asks participants to answer two scales per item: the degree to which the item is better in online or in person learning, and the degree to which the item is influential in their decision to choose online or in person learning. The survey will be pilot tested with 400 student participants. Confirmatory factor analysis will be performed on the results to test the fit of the factor structure on the new data. Each factor grouping will be evaluated for reliability to improve the survey for future use. With sufficient pilot testing and refinement, this future questionnaire can be one of the first of its kind with the ability to capture a detailed profile of students’ attitudes toward aspects of their learning experience. The results will enhance our ability to specifically target areas for improvement, and prioritize experiences that students weigh most heavily in their decision to enroll in online or in person learning.

Qualitative Content Analysis

Qualitative content analysis was conducted to understand the responses to the open-ended question in Section 3b, “Please list the factors that are most important to you when deciding whether to take a job that requires working remotely (your answers do not have to come from any factors that have already been listed in this survey so far).”

Coding of the responses was performed inductively based on latent content analysis, with themes developed during the coding process based on the content itself, rather than developing themes beforehand to describe the content (Elo & Kyngäs, 2008). Codes were defined as separate ideas within responses (e.g. a response of “flexibility, the ability to save money on commuting” would contain two units of meaning, separated by the comma). Out of the 456 participants who responded to the question, a total of 784 codes were identified.

Using qualitative coding software Quirkos, responses were independently coded by two coders - myself and an undergraduate assistant who had completed a course in qualitative methods. We then consolidated our work into separate reports to compare our creation of codes and themes. There was a high degree of consistency based on which individual codes were grouped in a theme together, though the themes themselves had different names. We met to discuss our code assignments and come to a consensus on the names of the themes. The assignment of codes was then revised by me to reflect any changes to where a code would be most appropriate. Any discrepancies in code assignment were either resolved through discussion or flagged for a third coder to review. Thirty responses were flagged for review by another undergraduate honours student. The student acted as a tiebreaker for the disputed items to determine the theme most appropriate for the given code.

The 7 emergent themes were Financial Needs, Flexibility, Personal Preference, Social Engagement, Life outside of work, Location, Job Nature. The final codebook can be found in Appendix B. Interestingly, several of the identified emergent themes were similar to extracted factors in Part 2. In particular, the content of the factors of Social Engagement, Resources, Life Outside of School, and Industriousness conceptually relate to the themes of some of the most common responses to the open-ended question. This provides some indication that, despite some

contextual differences like the additional consideration of salary, many of the same considerations that drive online learning preference also drive remote work preference. Additional causal research is required to determine the extent to which considerations and preferences for remote work may predict intentions to accept remote work, and the degree to which each consideration should be prioritized when designing remote work programs to attract applicants.

Discussion

Despite gaining popularity over the past two decades, online learning and remote work have generally struggled with higher dropout rates and lower job satisfaction, respectively. Our understanding of this disconnect between the growth of online modes and the attitudes of online workers has historically been stunted by several limitations., such as inconsistent personality measures, lacking the ability to compare online and in-person modes, and conflating preference with related criterion without directly measuring it. Additionally, past studies were typically limited to measuring samples that had already self-selected into online modes.

The widespread shift online due to COVID-19 created a large, novel population of individuals who were experiencing online learning and remote work for the first time. This study was conducted in response to the emergence of that population and achieved several objectives to better understand how individuals' personalities impact their preference for virtual modalities.

Firstly, this study confirmed that students' overall preference for online learning significantly correlates with their preference for future remote work. This suggests that students who have experienced online learning likely hold some of the same perceptions of remote work as they do with online learning, so their experiences with online learning may impact their future decisions about remote work opportunities. Future research on attitudes toward remote work can provide specific insight into the mechanisms and extent of such similarities, such as whether students value specific factors of online learning as equally as they value those factors in remote work. For example, is social connectivity just as important to people in work as it is in school? Additionally, at what point does the benefit of a higher salary outweigh the influence of less desirable factors? Investigating such questions will provide clarity for the ways that remote work

can be improved to address the needs of potential remote workers and attract desirable remote work applicants.

Secondly, this study found that the relationship between students' personalities correlate slightly with their overall preferences for online learning and remote work. Honesty-Humility, Emotionality, and Extraversion, each had a negative effect on students' preferences for online learning, and Honesty-Humility, Emotionality, Extraversion, and Openness to Experience each had a negative effect on students' preferences for future remote work. When online learning preference and remote work preference were separately regressed on the HEXACO factors, only Honesty-Humility and Extraversion significantly predicted both variables at an alpha of .01.

This findings of this study contradict those of Pavalache-Ille and Cocorada (Pavalache-Ilie & Cocorada, 2014), who found that high introversion was associated with participation in and preference for online learning. This study also supported the direct measurement of preference and its distinctiveness from satisfaction by contradicting the findings of Cohen and Baruth (2017) and Audet et al. (2021). While Cohen and Baruth found that Extraversion was not a significant predictor of satisfaction with online courses, this study revealed that Extraversion is a significant predictor of preference for in-person learning. Audet et al. (2021) found that none of the FFM traits predicted satisfaction with online courses, which is not reflected in the current findings of HEXACO traits that predict online learning preference.

It is curious that Openness to Experience reached significance in predicting remote work preference but not online learning preference. Perhaps for most students, remote work is still an unknown experience, and therefore would dissuade students who are low in Openness to Experience, whereas all students were forced to experience online learning for several years so whether they were "open" to the online learning experience or not was irrelevant by the time of

this study. Future studies that control for different levels of experience with online modalities may verify the existence of this discrepancy. For example, the Openness to Experience and preferences of students who have already completed online learning could be compared to those who have never completed online learning,

While Conscientiousness is consistently a strong predictor of various academic outcomes, including students' "impression" of online learning courses, this study did not find Conscientiousness to be a significant predictor of preferences for either online learning or remote work. Highly conscientious students tend to do well in their courses regardless of the mode of delivery. We cannot predict students' preference for a particular mode over the other based on their conscientiousness or performance if we only look at one mode without comparing both modes. This brings the conclusion and recommendations of Cohen and Baruth (2017) into question; the authors advocate for customizing online learning courses for students who are low in Conscientiousness so that they will be more satisfied with their online courses. However, if students low in conscientiousness are typically less satisfied with their courses regardless of delivery format, then the allocation of time and resources into redesigning online courses may reap a much smaller reward than the authors anticipate.

One clue into the true nature of Conscientiousness' relationship with online learning preference is the fact that students' subjective academic performance in online learning compared to in-person learning was a strong predictor of preference. This suggests that students' preferences may have less to do with whether Conscientiousness level in general was either a weakness or a strength in online learning during COVID-19 and instead depend on whether students ended up being successful in adapting their study habits according to the changes of online learning. Perhaps students who successfully adapted their techniques and methods for

completing coursework thrived online, more so than their peers who continued using the same strategies that used to work well for them in in-person learning, regardless of their Conscientiousness. Studying the interaction of students' Conscientiousness with their Openness to Experience on their performance and preference, as well as surveying them on the difference between their study habits pre- and post-switch to online, may provide more insight into who found more success adapting to online learning. Again, while this may actually be the case, the nature of only measuring overall preference in this study simply prevented that level of specificity about the relationship from being achieved. It is also possible that Conscientiousness may have always been irrelevant to students' preference, and that relationship had simply never been measured directly. In addition to advocating for the explicit measurement of preference rather than conflating with other measures, this study recognizes that in order to understand the relationship between students' personalities and preferences for online learning, the mechanisms behind the relationship must be explored more specifically.

Finally, this study took the first step in investigating the mechanisms of that relationship by conducting an EFA of 40 items related to higher education course enrollment that participants rated as being a better experience in either online or in-person learning. Six factors were extracted and represent the most relevant dimensions of the learning experience that influence students' preferences for learning modality. These were titled Social engagement, Industriousness, Resources, Counterproductivity, Individual Participation, and Life outside of school. The identification of these factors will guide the creation of a future assessment to elucidate why students prefer online or in-person learning and the most impactful factors in their decisions to enroll in either mode. By understanding the most important factors to students' course mode preferences, higher educational institutions can tailor online learning programs to

be more accommodating and attractive to a wider range of students, and address current challenges in the online learning experience. Prospective students can then be more confident when self-selecting into the learning mode of their choice.

Strengths and Limitations

This study made several key improvements and contributions to the literature. Notably, this study was among the first of its kind to utilize a sample of online learners who were not inherently biased in favor of online learning. Additionally, this study utilized HEXACO instead of a five factor model, which allowed the finding of Honesty-Humility correlating with preference for in-person learning to emerge. Also, this study measured perceptions of online learning compared to in-person learning, which resulted in findings that were different from similar studies that have otherwise assumed that their results were unique and representative of one mode.

This study also acknowledges several limitations. First, the timing of this study occurring during the COVID-19 pandemic represents a “double-edged sword” for the generalizability of the results. On the one hand, this study would not have had the ability to sample a diverse population of online learners if the COVID-19 pandemic had not caused a mass shift online. On the other hand, the general context of the ongoing pandemic may have had increased the stress or mental health struggles of the participants. While this study could not control for the occurrence of the pandemic itself, it did not include any mention of the pandemic until the end of the survey. This served to prevent invoking undue stress or negative feelings associated with the pandemic while participants completed the survey.

This study is also limited in its ability to draw direct conclusions about the importance of each factor in students’ decision making. Students rated each factor of the learning experience as

being better or worse within online learning, as well as their overall preference for online learning, but not the degree to which each factor is influential in their preference. Future research would be able to make a stronger causal claim for what drives students' enrollment decisions by measuring not only students' opinion about the individual factors but also the factors' importance to their preference.

There are of course many other factors besides personality that affect preference, such as personal needs, accessibility, and lifestyle. While this study was particularly interested in personality and cannot yet report causal relationships that make up students' enrollment decisions, the insight yielded by the EFA and qualitative analysis in this study provides guidance for the design of future studies that can dive deeper into the mechanisms behind students' learning mode preferences.

Notably, this sample was limited to one university in North America. The shift to online learning was abrupt and not standardized across universities, and therefore students across the world experienced online learning differently depending on their institution's unique implementation of online learning programs. Other higher educational institutions implemented online learning in their own way, influencing students' experiences with and attitudes toward the different learning modes. Differences in technological infrastructure, degrees of familiarity with other forms of online engagement, and political mandates regarding COVID-19 are some examples of variables that would impact implementation of online learning. Future studies could include students from multiple higher educational institutions and allow for comparison, as well as help institutions identify areas for improvement that may be more relevant to their program design. A nuanced understanding of students' attitudes toward online learning would facilitate a

tailored approach to improving the online learning experience and help increase the retention rate of online learners.

Conclusion

Online learning is rapidly expanding, especially following COVID-19, but it has a higher dropout rate than in-person learning. Remote work is also becoming more widely adapted by organizations due to benefits like a larger talent pool and fewer overhead costs. However, since individuals' personalities influence their attraction to online learning and remote work, organizations who do not consider the impact of personality on modality preference may be limited in their ability to achieve recruitment and performance goals. Understanding the individual differences of students' personalities and their preference for online learning is one step in elucidating what drives students' online course enrollment and drop-out intentions, as well as intention to accept remote work in the future. This understanding will help organizations improve their online or remote/hybrid programs to attract and retain happier, more effective students and employees.

Appendix A: Survey

(PART 1) Personality

60-question HEXACO-PI-R (Ashton & Lee, 2009).

Section 1.

Below you will find a series of statements about you. Please read each statement and decide how much you agree or disagree with that statement. Then select your response using the options to the right of each statement.
Please answer every statement, even if you are not completely sure of your response.
1 = strongly disagree
2 = disagree
3 = neutral (neither agree nor disagree)
4 = agree
5 = strongly agree

1. I would be quite bored by a visit to an art gallery.
2. I plan ahead and organize things, to avoid scrambling at the last minute.
3. I rarely hold a grudge, even against people who have badly wronged me.
4. I feel reasonably satisfied with myself overall.
5. I would feel afraid if I had to travel in bad weather conditions.
6. I wouldn't use flattery to get a raise or promotion at work, even if I thought it would succeed.
7. I'm interested in learning about the history and politics of other countries.
8. I often push myself very hard when trying to achieve a goal.
9. People sometimes tell me that I am too critical of others.
10. I rarely express my opinions in group meetings.
11. I sometimes can't help worrying about little things.
12. If I knew that I could never get caught, I would be willing to steal a million dollars.
13. I would enjoy creating a work of art, such as a novel, a song, or a painting.
14. When working on something, I don't pay much attention to small details.
15. People sometimes tell me that I'm too stubborn.
16. I prefer jobs that involve active social interaction to those that involve working alone.
17. When I suffer from a painful experience, I need someone to make me feel comfortable.
18. Having a lot of money is not especially important to me.
19. I think that paying attention to radical ideas is a waste of time.
20. I make decisions based on the feeling of the moment rather than on careful thought.
21. People think of me as someone who has a quick temper.
22. On most days, I feel cheerful and optimistic.
23. I feel like crying when I see other people crying.
24. I think that I am entitled to more respect than the average person is.
25. If I had the opportunity, I would like to attend a classical music concert.
26. When working, I sometimes have difficulties due to being disorganized.
27. My attitude toward people who have treated me badly is “forgive and forget”.

28. I feel that I am an unpopular person.
29. When it comes to physical danger, I am very fearful.
30. If I want something from someone, I will laugh at that person's worst jokes.
31. I've never really enjoyed looking through an encyclopedia.
32. I do only the minimum amount of work needed to get by.
33. I tend to be lenient in judging other people.
34. In social situations, I'm usually the one who makes the first move.
35. I worry a lot less than most people do.
36. I would never accept a bribe, even if it were very large.
37. People have often told me that I have a good imagination.
38. I always try to be accurate in my work, even at the expense of time.
39. I am usually quite flexible in my opinions when people disagree with me.
40. The first thing that I always do in a new place is to make friends.
41. I can handle difficult situations without needing emotional support from anyone else.
42. I would get a lot of pleasure from owning expensive luxury goods.
43. I like people who have unconventional views.
44. I make a lot of mistakes because I don't think before I act.
45. Most people tend to get angry more quickly than I do.
46. Most people are more upbeat and dynamic than I generally am.
47. I feel strong emotions when someone close to me is going away for a long time.
48. I want people to know that I am an important person of high status.
49. I don't think of myself as the artistic or creative type.
50. People often call me a perfectionist.
51. Even when people make a lot of mistakes, I rarely say anything negative.
52. I sometimes feel that I am a worthless person.
53. Even in an emergency I wouldn't feel like panicking.
54. I wouldn't pretend to like someone just to get that person to do favors for me.
55. I find it boring to discuss philosophy.
56. I prefer to do whatever comes to mind, rather than stick to a plan.
57. When people tell me that I'm wrong, my first reaction is to argue with them.
58. When I'm in a group of people, I'm often the one who speaks on behalf of the group.
59. I remain unemotional even in situations where most people get very sentimental.
60. I'd be tempted to use counterfeit money, if I were sure I could get away with it.

(PART 2) Online Learning Preferences

Section 2a.

The following list includes the pros and cons of online learning. Please indicate how you would evaluate your online learning experience with respect to each of these items *compared to* in-person learning.
That is, for each of the following items, to what extent online learning is worse or better than in-person learning? Indicate your response using the following scale.
-3 = Online learning is much worse
-2 = Online learning is worse
-1 = Online learning is a little bit worse

0 = No difference
1 = Online learning is a little bit better
2 = Online learning is somewhat better
3 = Online learning is much better

1. Work/life balance
2. Flexibility to complete schoolwork within my own schedule
3. Amount of free time (time spent doing non-schoolwork activities)
4. Overall financial savings (e.g. transportation/parking, food, school supplies, childcare, other costs related directly or indirectly to being enrolled in courses)
5. Overall financial value (i.e. getting "the most for your money" from your courses)
6. My overall sense of community
7. Opportunity to make new friends
8. Opportunity for social activities with other students
9. Opportunity for participating in extracurricular activities
10. Building professional rapport with instructor
11. Effective communication with instructors
12. Effective communication with other students
13. Engaging lectures (e.g., entertaining lecture delivery)
14. Collaboration experience with other students (e.g., working on team projects for courses, outside of class time)
15. Quality of class discussions (in class or through discussion boards)
16. Ease of participation within class activities (answering questions, discussions, completing polls)
17. Comfort as a presenter/speaker (e.g. controlling nervousness)
18. Logistics of easily delivering a presentation
19. Feeling engaged by other students' presentations, as an audience member
20. Evaluation fairness (i.e., exams and essays)
21. Turnaround time for grading
22. Ability to get my questions answered by instructors
23. Receiving needed accommodations (exam/paper extension, access to notes, etc.)
24. Access to the physical technological resources you need (i.e. Wi-fi, computer, hardware, work set up)
25. Technological difficulties/problems (i.e. Wi-fi, computer, hardware, work set up)
26. My motivation to learn
27. My discipline to complete my course work
28. My psychological health (Consider for example: work- or exam-related anxiety, stress, burnout)
29. My physical health (Consider for example: sedentary lifestyle, exercise habits, nutrition)
30. Feeling of accomplishment upon completing assignments/exams/projects
31. Feeling of accomplishment upon completing overall courses
32. Ease of getting away with cheating (i.e. engaging in plagiarism, using unauthorized materials)
33. My ability to resist the temptation to cheat (i.e. not engaging in plagiarism, not using unauthorized
materials)

34. Ease of taking advantage of group members (i.e. freeloading, not putting in fair share of work)
35. My ability to resist the temptation to freeload in group projects
36. My ability to get by with bare minimum effort in my course work
37. My consistency of attendance to scheduled lectures/labs (excluding legitimate absences)
38. My personal standards for the quality of work I submit
39. The actual quality of my work that I submit (assignments, papers, exams)
40. Overall learning quality / quantity

Section 2b.

Overall, which form of learning do you prefer more: online learning or in-person learning?
-3 = I prefer online learning much more
-2 = I prefer online learning more
-1 = I prefer online learning a little bit more
0 = I have no preference
1 = I prefer in-person learning a little bit more
2 = I prefer in-person learning more
3 = I prefer in-person learning much more

(PART 3) Future Remote Work Preferences

Section 3a.

Now imagine you are considering full-time work after University for a position that could be done either remotely or in-person. The following statements refer to your preferences for remote work and in-person work.

"Remote work" refers to the practice of working online, outside of a traditional in-person office space. One might work from their computer at home or from a secondary location like a library, café, or personally rented office space. Note that remote work is on a continuum, and can range from fully remote (spending 100% of your working time away from a traditional office space) to occasionally remote (spending most of your time at a traditional office space). For the purposes of this survey, assume that remote work means 80% or more of your time is spent working remotely.

For the following statements, indicate how much you agree or disagree with that statement. To what extent do you agree with the following statements?

1 = strongly disagree
2 = disagree
3 = neutral (neither agree nor disagree)
4 = agree
5 = strongly agree

1. I have read the instructions and understand that I am now thinking about future, hypothetical full-time employment (please select “strongly agree” for this statement).
2. I would accept an offer of remote work from my employer over an offer to work in-person.
3. Overall, the positives of remote work outweigh the negatives.
4. Remote work might be good for some people but not for me.
5. Ideally, I would like to split my week equally by working remotely on some days and working in-person on some days.
6. I’d like to work remotely most days of the week, but be in-person a couple of days per week.
7. If I was offered a similar position at another organization where I could work remotely at least some days, and my current organization only worked in-person, I would leave my current organization to take the semi-remote offer.
8. If my organization switched to only working remotely, and I had an offer to work in-person for a similar position at another organization, I would leave my current organization to take the in person offer.
9. I would never want to work remotely for my job.
10. I would prefer to work fully remotely over working in-person.

Section 3b.

Please list the factors that are most important to you when deciding whether to take a job that requires working remotely (your answers do not have to come from any factors that have already been listed in this survey so far):
[Open-ended response box]

(PART 4) Additional Items

Section 4a.

1. What is your age?
   [drop down list: 1-100]
2. What is your gender identity?
   [multiple choice: woman, man, transgender woman, transgender man, gender variant/non-conforming, not listed, prefer not to answer]
3. What is your sex?
   [multiple choice: female, male, transsexual female, transsexual male, intersex, not listed, prefer not to disclose]

Section 4b.

1. Are you currently employed?
   [multiple choice: yes, no]
2. (If yes to question 1) What is the modality of your current job?
   [multiple choice: In-person work, Partially remote (some days of the week in person), Fully remote (no days in person)]
3. How many months of employment history do you have in the following modalities?
   [number entry for each: In-person work, Partially remote (some days of the week in person), Fully remote (no days in person)]

Section 4c.

1. COVID-related anxiety: To what extent does your personal fear about COVID (on behalf of yourself, a loved one, the healthcare system, etc.) play a role in your course selection decisions (i.e. whether to take online versus in-person courses)?
   [multiple choice: none, a little, some, a lot]
2. Significant personal factors: To what extent do factors such as financial strain, extenuating family obligations, parenthood, etc. play a role in your course selection decisions?
   [multiple choice: none, a little, some, a lot]
3. Academic performance: How would you rate your academic performance in your most recent online courses (compared to your most recent in-person courses)? "Compared to my in-person courses, my performance in my online courses has been ____"
   [multiple choice: much worse, somewhat worse, slightly worse, no different, slightly better, somewhat better, much better]
4. Online learning experience: In your most recent semester (Fall 2021), **when you initially selected your courses**, how many of your selections were scheduled to be delivered online asynchronously (with no specified lecture time, self-paced), online synchronously (with a specified lecture time), hybrid (some lectures/labs in person, some online), and in-person?
   [number entry for each: online asynchronously, online synchronously, hybrid, in-person list]
5. Online learning experience: In your most recent semester (Fall 2021), how many of your courses were ultimately delivered as primarily (for most of the semester) online asynchronously (with no specified lecture time, self-paced), online synchronously (with a specified lecture time), hybrid (some lectures/labs in person, some online), and in-person?
   [number entry for each: online asynchronously, online synchronously, hybrid, in-person list]
6. Online learning experience: In your current semester (Winter 2022), **when you initially selected your courses**, how many of your selections were scheduled to be delivered online asynchronously (with no specified lecture time, self-paced), online synchronously (with a specified lecture time), hybrid (some lectures/labs in person, some online), and in-person?
   [number entry for each: online asynchronously, online synchronously, hybrid, in-person list]
7. Online learning experience: In your current semester (Winter 2022), how many of your courses are currently delivered online asynchronously (with no specified lecture time, self- paced), online synchronously (with a specified lecture time), hybrid (some lectures/labs in person, some online), and in-person?
   [number entry for each: online asynchronously, online synchronously, hybrid, in-person list]

Appendix B: Qualitative Responses Codebook

Emergent Themes of Factors Impacting Remote Job Acceptance and Example Codes

Emergent Theme	Description	Example Codes
Financial Needs	The pay and benefits provided by the job.	"Pay" "Wages" "Benefits" "Financial situation" "Expenses" "Income" "Cost of living"
Flexibility	Having the ability to customize one's schedule and/or location of remote working.	"Flexibility" "Option to stay home" "Option of in person is available" "Flexible hours"
Personal Preference	Consideration for personal characteristics affecting preference, comfort, and compatibility with the job.	"COVID-19" "My motivation" "Mental and physical health" "My goals" "If the job is engaging" "Interest" "Disability accommodation"
Social Engagement	Ability to socialize within and outside of work.	"Friends and social life" "Social aspect" "Community" "Employee interaction"
Life outside of work	Work-life balance that accommodates caretaking obligations and free time away from work.	"Kids" "Family" "Home life" "Work-life balance" "Childcare" "Taking care of pets" "Amount of free time"
Location	Considerations such as moving or commuting.	"Travel time" "Commute"

		“Expenses from transportation” “Whether I have to move”
Job Nature	Characteristics of the job and company culture.	“Hours of work” “Work environment” “Resource availability” “Advancement” “Amount of teamwork” “Level of supervision” “Amount of work” “Frequency of meetings” “Company culture” ”Deadlines”

Appendix C: Plan for Next Steps in Scale Development

(Boateng, Neilands, Frongillo, Melgar-Quiñonez, & Young, 2018; Carpenter, 2018)

	Theme	Description
1	Social Engagement	Experience of engaging with other students and professors inside and outside of the classroom
2	Industriousness	Personal qualities and behaviors to produce high-quality work
3	Resources	Having the resources and accommodations, both tangible and intangible, to complete coursework
4	Counterproductivity	Ease and temptation to engage in counterproductive behaviors
5	Individual Participation	Ease and satisfaction with class activities completed individually, such as presenting
6	Life Outside of School	Ability to customize the time and resources dedicated to life outside of school

The contributions of the present study pave the way for the creation of a novel assessment of attitudes toward online and in-person learning. The planned scale aims to improve upon existing measures by including both online and in-person modes and including items that represent a wider breadth of the learning experience. Together, these improvements will allow for comparison between the modes and capture a richer view of learning mode preference. The themes extracted by the EFA in the present study (described above) will guide the item generation for this new scale. An overview of the scale creation plan is presented below.

Step 1. Generate items

A total of 60 items will be written, split equally across the six dimensions. The content of these items will be informed primarily by the results of the EFA in the present study. The qualitative responses gathered in the present study, as well as review of scales used in other studies that measured attitudes toward coursework (e.g. Audet et al., 2021; Paechter & Maier, 2010), will also be referenced during item generation. Similar to the approach seen in Organizational Citizenship Behavior research (Rioux & Penner, 2001) in which items describe both a behavior and motive for the behavior, the items in this scale will be formatted to include both an attitude (preferring online or in-person learning) and a justification (e.g. "because it is easier to build rapport with the professor"). Responses will be gathered on a 5-point scale from 1 (strongly disagree) to 5 (strongly agree). Sample items reflecting each dimension include:

1. "I prefer in-person learning because it is easier to become friends with my classmates."
2. "I prefer in-person learning because being in the classroom helps me focus more on the lecture."
3. "I prefer in-person learning because I can get my questions answered more quickly."
4. "I prefer online learning because I can look up the answer to a test question without getting caught."
5. "I prefer online learning because it makes it easier to give class presentations."
6. "I prefer online learning because it allows me to do my work wherever I like."

Step 2. Evaluate initial content validity of generated items ("Meaning analysis"; Chaffee, 1991)

The preliminary items will be pre-tested with a group of 5-10 subject matter experts (SMEs). The SMEs will be provided with the six dimension descriptions and the purpose of the scale – to measure the relationship between elements of the learning experience and overall learning mode preference.

The SMEs will rate the validity of each item in terms of relevance and clarity (on a scale of 1 to 4, plus a free-response box to provide feedback).They will also be invited to suggest possible items that they think should be included. Using the SMEs' item validity ratings, the content validity ratio will be calculated for each item (i.e. the proportion of SMEs who rated item 3 or higher in relevance and clarity out of total number of SMEs). Any items that have a content validity ratio of less than .75 on either relevance or clarity will be flagged for review.

Finally, all items flagged for low content validity will be rephrased or replaced. The open-ended feedback provided by SMEs will be used to guide these revisions when applicable. The SME review process will be repeated for the rewritten items until high content validity is achieved for all items.

Step 3. Administer survey

Using a new sample of 400 participants, the scale will be administered through an online survey. In addition to the 60 scale items, participants will be asked to rate their general attitudes toward either learning mode, including:

Overall preference for either learning mode (on a scale of -3 (*I prefer in-person learning much more*) to 3 (*I prefer online learning much more*));

Choice between an online or in-person format for their favorite course this semester (on a scale of -3 (I would definitely choose an in-person format) to 3 (I would definitely choose an online format));

Intention to enroll in online or in-person courses in the future (on a scale of -3 (I would enroll in all in-person courses if I could) to 3 (I would enroll in all online courses if I could)).

Step 4. Select final items

After data collection, several criteria will be considered to guide the selection of 5 items per dimension (30 items total).

The descriptive statistics (mean, standard deviation, and item-total correlation) will be computed for each item. Items with an anomalous mean that is not close to the midpoint (3 on the scale of 1-5), and that has a small standard deviation, or low item-total correlation (<.30) will be considered for deletion. The content of the items will also be considered when selecting items to ensure that the final list of items represents the breadth of the criterion of the scales.

After a final set of items is chosen, internal consistency reliabilities will be computed for the final scales. If the reliabilities fall in the acceptable range (approximating .70), then the scales will demonstrate internal consistency and we will proceed with a Confirmatory Factor Analysis to examine the hypothesized six-factor solution provide a good fit to the data. The goodness of fit indices of Root Mean Square Error of Approximation (RMSEA ≤ 0.06), Tucker Lewis Index (TLI ≥ 0.95), and Comparative Fit Index (CFI ≥ 0.95) will be evaluated. If the model underperforms according to fit indices, additional items will be identified for removal according to the strength of their factor loadings. Items with weak factor loadings (<.30) will be replaced and the revised scale will be re-tested with a new sample. The correlations between the dimensions will be evaluated to ensure that they are distinct from one another. If any dimensions are highly correlated, collapsing those dimension to a single dimension will be considered.
The cross-sectional construct validity of the scale will be assessed by analyzing its convergence and divergence with the items selected in Step 4.

Step 5. Conduct a construct validity study

The scale will then be administered to a new sample of 400 participants to evaluate construct validity. In addition to the final set of scales selected in Step 4, participants in the new sample will also be asked to rate the following variables (Some of the scales in the list will be constructed ad hoc for this study):

1. Choice between online and in-person learning for their favorite course;
2. Intention to enroll in online or in-person courses in the future;
3. Satisfaction with online and in-person courses;
4. Perceived value for money according to learning mode;
5. Value of having a social life;
6. Importance of being evaluated fairly by course instructors;
7. Average stress level during semester;
8. Enjoyment of trying new hobbies;
9. Enjoy working on a team;
10. Competitiveness with myself;
11. Personal tech savviness;
12. HEXACO Personality
13. Interest in watching sports.

The scale will demonstrate convergent validity if the scale items correlate significantly and positively (<.50) with criterions 1-9, and demonstrate divergent validity if the items have little to no correlation with criterion 10.

Variables 1-4 will be used to validate the scale as a whole and were selected because they are theoretically related to learning mode preference and could reasonably be measured using the same set of scales that are being used to measure learning mode preference.

Variables 5-11 will be used to validate the convergent validity of the scale according to the 6 dimensions in the scale. Each of the variables is conceptually related to one dimension. For example, “Value of having social life,” is expected to relate to dimension 1 (Social Engagement). The HEXACO personality inventory (variable 12) will be used to further evaluate the construct validity of the new scales. Since some dimensions also correlate with one or more HEXACO

traits, it is expected that the variable will also correlate with an associated trait. Based on the results of the present research Extraversion is hypothesized to correlate primarily with dimension 1 (Social Engagement), because students high in Extraversion are more likely to favor a mode facilitating their social interaction. Honesty-Humility and Conscientiousness are expected to negatively correlate with dimension 4 (Counter productiveness), because that students high in those traits are more likely to disfavor a learning mode on the basis of it providing more opportunities for cheating. The variable for measuring divergent validity of the scale (number 13) was selected because it is theoretically unrelated to learning mode preference and any of its dimensions, and could not be measured using the same set of scales.

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